

Technology Stack

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	2 Marks

The table below details the technologies chosen for each layer of the Rising Waters architecture.

S. No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS	Simple, lightweight form-based UI that needs no additional frontend framework.
2	Backend / Server-Side	Python, Flask	Lightweight Python web framework, ideal for quickly serving a trained ML model.
3	Machine Learning	Scikit-learn (Random Forest Classifier)	High accuracy on tabular weather data, resistant to overfitting.
4	Data Processing	Pandas, NumPy	Efficient loading and transformation of structured rainfall/weather data.
5	Model Serialization	Joblib	Fast, reliable saving and loading of the trained scikit-learn model.
6	Data Storage	Excel dataset (flood_dataset.xlsx) via OpenPyXL	Dataset is static and tabular; a full database is unnecessary for this scope.
7	Development Environment	Jupyter Notebook, VS Code	Convenient for exploratory data analysis and iterative model development.
8	Version Control & CI/CD	Git & GitHub	Enables version tracking and public sharing of the codebase.
9	Cloud / Hosting / Deployment	Planned: Render / Heroku (not yet deployed)	Free-tier hosting suitable for a lightweight Flask demo application.
10	Third-Party APIs / Other Tools	Planned: OpenWeatherMap API	Would enable real-time weather data instead of manual input.